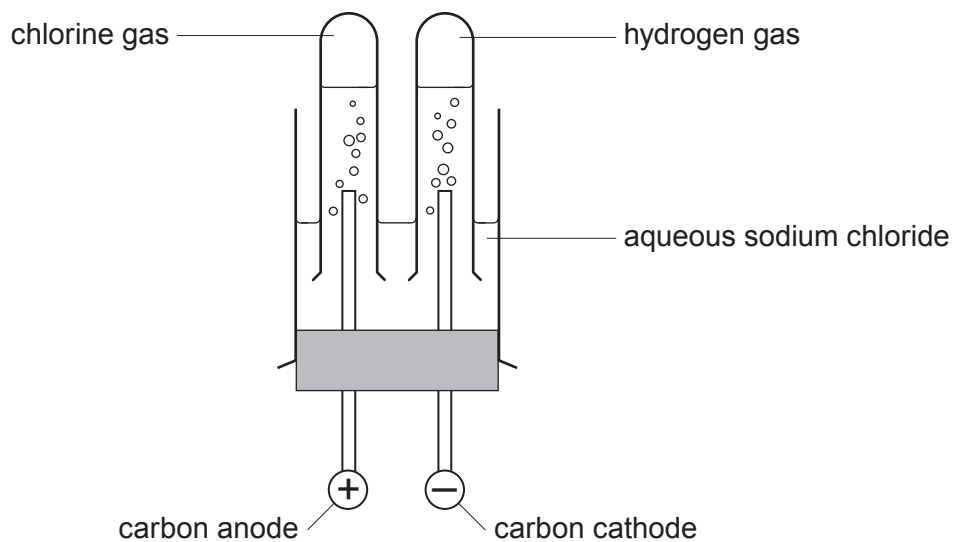


11. The diagram shows the laboratory apparatus that can be used to model the manufacture of aqueous sodium hydroxide (NaOH), chlorine (Cl_2) and hydrogen (H_2) from aqueous sodium chloride, NaCl .



Aqueous sodium chloride contains the ions Na^+ , Cl^- , OH^- and H^+ .



Explain, in terms of ions, the formation of each of the **three** products. Include equations to support your answer. [6 QER]

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END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

1 H Hydrogen 1										4 He Helium 2									

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number